

SPPA-MVFit Supplement: Protocol, Reproduction, and Secondary Evidence

Purpose

This supplement records the confirmatory protocol path and reproduction commands, the external neural-wave protocol, the open-label probe, the real-image probes and replay audits, the role-preservation diagnostic, the descriptor and Unreal runtime tests, and the full tables behind the figures of the main paper.

Primary endpoint (held-out)

- Estimand: clean 64-cubed voxel IoU of `sppa_mvfit` minus `generic_mvfit`. Units: 240 source actors (12 family×stratum cells, equal strata).
- Decision: stratified bootstrap 95% CI lower bound $> +0.030$. Result: mean $\Delta = 0.190$, CI $[0.181, 0.199]$, **H1 PASS**.
- Strata: CSG-ID 0.209, implicit-ODD 0.172 (both positive). Provenance: `synthetic_geometry` only.

Reproduction commands

Run from the git repository root (or the `paper_semantic_proxy_3d` junction directory inside it). Do not re-optimize the held-out methods after inspection.

Table S1: Minimal reproduction checks for the primary claim.

Command	Purpose	Expected
<code>python-mpytestpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/tests/test_contract.py</code>	Shared builder, budget, no cross-imports, split balance.	7 passed
<code>pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/verify_package.py--development</code>	Development package integrity.	PASS
<code>pythonpaper_semantic_proxy_3d/tools/reproduce_sppa_mvfit_paper.py--strict</code>	Hash/gate checks for the submission snapshot.	0 blockers

Held-out generation/evaluation commands exist in `reproducibility/sppa_mvfit/README.md` and require the protocol PASS record, pretest freeze, and NIST-bound seed manifest. Re-running method optimization for a better score is forbidden by protocol. Main-paper tables are regenerated from the released `raw_metrics.csv` by `export_paper_tables.py`.

Protocol bookkeeping and artifact integrity

The margin (+0.030), the power analysis, and the observation conditions were fixed in writing before evaluation (protocol Amendment 01). Seeds were derived from a NIST Randomness Beacon pulse after a pre-test freeze and a written triple-role protocol release (Amendment 03); the external neural wave was registered as Amendment 05 before any measurement. SHA-256 manifests for the pretest freeze, the protocol PASS record, the prediction binary, and the raw metrics are stored with the artifact package under `reproducibility/sppa_mvfit/`.

Claim boundaries

Supported: family-conditioned occupancy gain under synthetic calibrated silhouettes; shared generator; optional runtime contract evidence. Not supported: measured flight, real-UAV 3D ground-truth accuracy, operator benefit, universal open-set geometry, neural image-to-3D SOTA ranking, or transfer of the preregistered occupancy margin to external real meshes (not replicated at $n=52$ in the post-hoc sanity check, main-text Section 4.7). Real-image figures

in this supplement remain qualitative probes without 3D ground truth and use declared replay assumptions where telemetry is shown.

S.1 Representation Design Space

The relevant comparison is not a single “3D representation” baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table S2 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools.

Table S2: Operational representation policy for semantic detections.

Situation	Preferred representation	Reason	Boundary
Matched project asset exists	Curated asset backend	best appearance and lower measured cost in the cow/tower subset	asset library must match class and scale
Known semantic family, no asset	SPPA primitive actor	persistent, local, role-labeled 3D proxy from label/pose input	finite reviewed archetypes
Unknown or adversarial label	SPPA fallback volume	avoids inventing a specific object from weak semantics	not visually specific
Clean crop plus offline budget	Image-to-3D or asset authoring	richer mesh can be generated outside the flight update loop	no per-frame track contract
Planning or occupancy task	Occupancy / ESDF / map layer	represents navigable space and collision evidence	not a role-labeled display actor

S.2 External Neural Wave: Protocol Details and Exclusions

Protocol. Sixty held-out test actors were selected deterministically (five per morphology-by-stratum cell, lexicographic `case_id` order). The measured generators were TripoSR [1] (`stabilityai/TripoSR`) and Hunyuan3D-2mini-turbo [2] (`tencent/Hunyuan3D-2mini`, checkpoint `hunyuan3d-dit-v2-mini-turbo`, octree 380, 5 sampling steps, FlashVDM). Each neural generator received two prespecified inputs: (a) a clean-crop shaded oblique render of the source actor, the generator’s best input condition, and (b) the actual 96×96 top telemetry mask as the input image, the closest visual analog of the channel SPPA consumes. Outputs were aligned to ground truth by a frozen frame convention (a one-time 48-candidate signed-permutation search fitted on twelve calibration cases; any selection bias this introduces can only inflate neural IoU, so the direction is conservative for SPPA), uniform scale to the ground-truth bounding box, and ground-truth bounding-box center and yaw frame; no per-case tuning was allowed. Voxel IoU was measured at 64^3 with the same voxelizer and WORLD frame. Two hard crashes (2/240 runs, Hunyuan3D oblique condition) are reported as exclusions; SF3D (install/build failure under the Amendment 05 Python 3.12 stack, after an earlier Python 3.10 attempt produced no benchmark event before a 20-minute timeout), SPAR3D (gated weights), and TRELLIS.2 (no Windows environment) remain documented environment-level exclusions, so no measured quality result exists for them in either attempt. SPPA-MVFit, Generic-MVFit, and visual-hull rows are the existing confirmatory clean-condition results on the same sixty cases.

Table S3: Measured external neural wave (Amendment 05, secondary). Mean voxel IoU at 64^3 on the same sixty held-out cases (Hunyuan3D (a): $n=58$ after the two reported hard crashes), with parenthesized paired Δ IoU versus SPPA-MVFit. Triangles and payload are mesh-output means; generation time is warm per-case wall time on the same RTX 5090; VRAM is uncapped peak CUDA allocation. SPPA rows are CPU-only and their payload is the update descriptor, not a mesh. Per-case values: `benchmarks/results/sppa_neural_external_wave.json`.

Method	Input	IoU	Triangles	ms	VRAM (MB)	Payload
SPPA-MVFit (ours)	masks top+side	0.561	536	9.2	–	1.45 kB
Generic-MVFit	same masks	0.364 (-0.197)	1,280	11.4	–	1.43 kB
Visual hull	same masks	0.516 (-0.045)	51,432	0.2	–	32.8 kB
TripoSR (a)	clean render	0.128 (-0.433)	28,145	384.1	1,869	1.47 MB
TripoSR (b)	top mask	0.231 (-0.331)	46,053	379.4	1,870	2.45 MB
Hunyuan3D-2m (a)	clean render	0.157 (-0.403)	686,733	1,350.9	4,591	10.5 MB
Hunyuan3D-2m (b)	top mask	0.171 (-0.390)	3,089,274	1,915.5	4,747	46.5 MB

S.3 Verifier-Gated Open-Label Probe

We evaluate the open-label route with 20 deliberately awkward labels that were not selected as the original paper demo classes. The set includes heavy equipment, vertical structures, containers, UAV-like objects, agricultural objects, and unsafe visual-artifact contexts. Table S4 reports the current verifier-gated recipe implementation. The metric and tag-only conditions use the same labels; metric inputs provide declared dimensions, while tag-only inputs use template priors.

Table S4: Uncached open-label SPPA probe. Family/proxy means non-fallback runtime output. Verified recipes are accepted only after deterministic checks over required roles, fallback-role absence, primitive budget, and dimensions.

Condition	Fallback	Family/proxy	Verified recipes	Critical mismatches
Verifier-gated recipes, metric	4/20	16/20	8/20	0
Verifier-gated recipes, tag-only	4/20	16/20	8/20	0

The remaining fallback cases in this probe are intentional: **dog statue**, **toy tractor**, **worker helmet**, and **cow silhouette** are visual-artifact or object-part contexts. This is a stronger claim than pure fallback but still narrower than arbitrary text-to-3D: SPPA accepts a generated proxy only when the recipe contract passes.

Quadruped profiles and open-label admission. For an animal-like object, SPPA does not infer anatomy from first principles: a label such as **cow**, **horse**, **dog**, or **quadruped** selects a morphology profile inside the same quadruped generator (torso proportions, head/neck placement, legs, tail), and the emitted proxy remains a low-polygon part graph, not a biological reconstruction. For a tractor or tractor-trailer the exact taxonomic class is not required: **vehicle**, **agricultural vehicle**, **truck**, or **trailer** normalize to a farm, heavy, or generic vehicle proxy. Weak or adversarial labels prefer the conservative fallback volume over an invented object.

S.4 Real-Image Probes, Replay, and Visual Evidence

Four real-image probes (cyclist, electrical tower, tractor, and tractor with trailer) exercise the dual-input contract of the main paper (Section 3.2). These user-supplied crops are real input evidence, not 3D ground truth. In the image path, a YOLOE-26s open-vocabulary detector reads the real image and produces approximate semantic evidence; in the tag-only path, SPPA receives a semantic tag/text input without using the image crop. In both cases the semantic text is routed through a SPPA normalizer before proxy selection, so the pipeline does not depend on YOLOE producing an exact object name. Table S5 summarizes the four probes.

Table S5: Real-image probe summary (audits in Tables S6–S9). YOLOE evidence is detector output, not ground truth. Raw dimensions come from mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the constraint-fused proxy dimensions of the observation-fusion audit; wall time is the detector+observation route proxy build. No 3D ground truth exists for these images, so no IoU is reported here.

Case	YOLOE evidence (conf.)	SPPA family	Raw dims (m)	SPPA dims (m)	Tris	Wall (ms)
Cyclist	person + motorcycle (0.40)	biker	$3.59 \times 1.77 \times 1.85$	$1.85 \times 0.73 \times 1.85$	1,036	3.528
Tower	electric pylon (0.46)	vertical_structure	$12.77 \times 2.89 \times 28.00$	$5.60 \times 2.89 \times 28.00$	270	1.372
Tractor	agricultural vehicle (0.52)	farm_vehicle	$4.95 \times 4.32 \times 2.60$	$4.52 \times 2.38 \times 2.60$	880	2.477
Tractor-trailer	vehicle (0.47)	articulated_vehicle	$16.85 \times 6.21 \times 3.40$	$12.11 \times 3.76 \times 3.40$	1,904	2.963

Table S6: Real-image SPPA observation and replay audit per probe. Raw dimensions come from YOLOE mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the constraint-fused proxy dimensions. Telemetry and height priors are declared replay parameters, so position and scale are scenario-relative, not measured ground truth.

Case	YOLOE evidence (conf.)	Replay proxy	Raw L/W/H (m)	SPPA L/W/H (m)	Gate	Local pose (m)	Tris
biker	person + motorcycle (0.40)	biker	$3.59 \times 1.77 \times 1.85$	$1.85 \times 0.73 \times 1.85$	soft low-conf fusion	$N = -3.67$, $E = -1.75$	1036
tower	electric pylon (0.46)	vertical_structure	$12.77 \times 2.89 \times 28.00$	$5.60 \times 2.89 \times 28.00$	height-only fusion	$N = -1.72$, $E = 0.56$	270
tractor	agricultural vehicle (0.52)	farm_vehicle	$4.95 \times 4.32 \times 2.60$	$4.52 \times 2.38 \times 2.60$	soft aspect fusion	$N = -8.45$, $E = -10.96$	880
tractor_trailer	vehicle (0.47)	heavy_vehicle	$16.85 \times 6.21 \times 3.40$	$12.11 \times 3.76 \times 3.40$	soft low-conf fusion	$N = -2.38$, $E = -7.61$	1904

Table S7: Real-probe semantic normalization and detector-plus-observation refinement (fusion of the normalization table and the refinement audit). The image-side detector evidence is not treated as exact ground truth; SPPA maps it to a proxy family, optionally refining weak detector labels when metric observation supports a safer conservative shape. The refinement route does not use the reviewed tag: it starts from YOLOE detector text and metric replay evidence; ms and Tris are its wall time and proxy size.

Probe	YOLOE image evidence	Detector-only family	Detector+observation family	SPPA reviewed text	Refinement rule	ms	Tris
Cyclist road image	person + motorcycle	two_wheeled_rider → biker	two_wheeled_rider → biker	biker	person+two-wheel	3.528	1036
Mountain tower image	electric pylon	power_tower → vertical_structure	power_tower → vertical_structure	tower	power infrastructure	1.372	270
Mountain tractor image	agricultural vehicle	farm_vehicle → farm_vehicle	farm_vehicle → farm_vehicle	tractor	farm vehicle	2.477	936
Tractor with trailer image	vehicle	generic_vehicle → heavy_vehicle	articulated_vehicle → articulated_vehicle	tractor+trailer	long-footprint metric refinement	2.963	1848

The dual-input benchmark of Table S8 freezes this two-path contract as an explicit artifact; it remains separate from any visual image-to-3D leaderboard because no 3D reference meshes exist for these real images.

Table S8: Frozen real-image and tag/text dual-input benchmark for the bounded SPPA systems claim. The image track uses YOLOE detector evidence and crops for image-to-3D baselines; SPPA additionally reports an observation-refined detector family when geometry supports a safer proxy. The tag/text track uses the reviewed word or phrase for SPPA and text-conditioned baselines. This is not a full visual image-to-3D leaderboard.

Case	YOLOE image input	SPPA family path	Image-track methods	Tag/text methods
biker	person + motorcycle (0.40)	detector: two_wheeled_rider → biker ; obs: two_wheeled_rider → biker ; text: biker → biker	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tower	electric pylon (0.46)	detector: power_tower → vertical_structure ; obs: power_tower → vertical_structure ; text: tower → tower	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor	agricultural vehicle (0.52)	detector: farm_vehicle → farm_vehicle ; obs: farm_vehicle → farm_vehicle ; text: tractor → tractor	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor+trailer	vehicle (0.47)	detector: generic_vehicle → heavy_vehicle ; obs: articulated_vehicle → articulated_vehicle ; text: tractor_trailer → tractor_trailer	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA

Figure S1 and Table S9 isolate the same SPPA generator under three input contracts: text-only (reviewed phrase), detector+metric (YOLOE text plus declared replay scale), and visual (agnostic image-space cues that may only condition existing roles). Visual evidence causes a measured geometry delta in all four cases at an additional cost of 20–40 triangles.

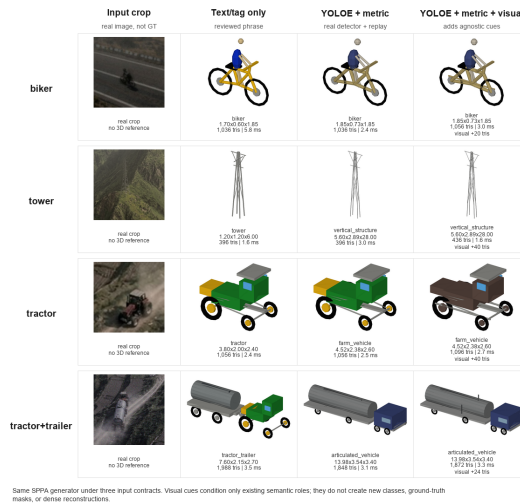


Figure S1: SPPA input-mode ablation on the four real-image probes: text-only, YOLOE+metric, and YOLOE+metric+visual contracts through the same deterministic generator. An ablation of SPPA evidence handling, not ground-truth 3D evaluation.

Table S9: SPPA input-mode comparison with bounded evidence-channel coverage. All rows call the same deterministic generator. Ch. counts the audited evidence channels consumed (semantic normalization, metric replay, visual role cues, yaw gating, observed material color); it is not a 3D quality score. All rows pass the lightweight runtime budget.

Case	Mode	Semantic	Dims L/W/H (m)	Visual tris	Δ vs metric	Ch.	Wall (ms)	Tris
biker	text	biker	1.70/0.60/1.85	-	yes	1	5.8	1036
biker	det+metric	biker	1.85/0.73/1.85	-	no	2	2.4	1036
biker	det+metric+visual	biker	1.85/0.73/1.85	20	yes	4	3.0	1056
tower	text	tower	1.20/1.20/6.00	-	yes	1	1.6	396
tower	det+metric	vertical	5.60/2.89/28.00	-	no	2	3.0	396
tower	det+metric+visual	vertical	5.60/2.89/28.00	40	yes	4	1.6	436
tractor	text	tractor	3.80/2.00/2.40	-	yes	1	2.4	1056
tractor	det+metric	farm	4.52/2.38/2.60	-	no	2	2.5	1056
tractor	det+metric+visual	farm	4.52/2.38/2.60	40	yes	5	2.7	1096
tractor+trailer	text	tractor+trailer	7.60/2.15/2.70	-	yes	1	3.5	1988
tractor+trailer	det+metric	articulated	13.98/3.54/3.40	-	no	2	3.1	1848
tractor+trailer	det+metric+visual	articulated	13.98/3.54/3.40	24	yes	4	3.3	1872

The refinement route of Table S7 does not use the reviewed tag: in the tractor-with-trailer case the system refines `vehicle` only to `articulated_vehicle`, never to the reviewed `tractor_trailer` class.

Metric replay and replay verifier. The metric proxy replay injects declared UAV pose, AGL, camera mount, and family height priors as scenario parameters; it is a real-image test of the observation-to-descriptor path, not a measured flight-localization result. A replay verifier checks the native-mask path: the selected YOLOE detections include native mask polygons with 12, 35, 33, and 54 points, projected into candidate oriented footprints before the constraint-fusion gate. This verifies the detector-evidence data path, not mask correctness against ground truth.

Visual bridge and connector regressions. An experimental agnostic image-space probe extracts generic cues (round-pair, line-structure, principal-axis, mask-envelope) from real crops and unlabeled masks; the descriptor consumes them only in a role-conditioned way, without creating new classes or topology. Connector regressions guard against disconnected-part proxies, and an anti-shortcut battery mutates labels, case identities, mask ordering, side channels, mirror transforms, and photometric variants; primary decisions pass in every row. These are plumbing checks, not evidence of universal primitive recovery; full tables and grids are retained in the artifact package (`benchmarks/results/`).

S.5 Role-Preservation Diagnostic

The implemented diagnostic asks whether SPPA preserves semantic part roles while adapting dimensions. For four synthetic vehicle variants, median dimension error was 0.3926 for fixed templates, 0.0000 for trivial global scaling, and 0.0223 for SPPA parametric vehicles; global scaling fits the outer box by deforming all parts together and is not a win condition. The supported claim is narrower: role-assigned parts can follow different update rules. A short/long truck invariance check makes this concrete: increasing vehicle length from 5.2 m to 8.2 m kept cab and tire scale deltas at 0.0, increased cargo length by 2.925 m, and increased tire count from 6 to 8. A calibrated-mask variant over 64 synthetic rectangle masks exercises the same path from observed footprint evidence (contract regression, artifact package).

S.6 Controlled Contract Regressions (Note)

Two controlled benchmarks verified the observation-to-descriptor plumbing as round-trip regressions under a known synthetic camera: a six-case contract benchmark (label-only prior versus metric-observation path) and a 162-mask bbox-versus-silhouette projection benchmark. Against self-generated synthetic references the metric paths recover the supplied dimensions exactly; these are scale-adaptation plumbing checks, not accuracy results on real detector masks, and the tables are retained in the artifact package (`benchmarks/results/`).

S.7 Superseded Substitutability Stress Test (Note)

An early substitutability stress comparison (July 2026; six ad-hoc local object cases; Shap-E, Point-E, TripoSR, and Hunyuan3D-2mini Turbo under a constrained local profile) is superseded for external-generator evidence by the measured neural wave of Section S.2 and is retained only in the artifact package. Its single durable observation is qualitative: generators that receive appropriate visual evidence produce richer meshes, at GPU cost and without a track-update contract.

S.8 Descriptor and Unreal Runtime Tests

In the descriptor/update replay benchmark, exact and keyword labels resolved to known finite archetypes and unsupported labels produced unknown fallback descriptors. Descriptor build with parts over 81,568 threshold-expanded rows had P50/P95/P99 of 391.8/515.0/671.2 μ s (maximum observation 360.050 ms); scheduler decisions had P50/P95/P99 of 9.2/14.2/19.2 μ s, and update-packet builds of 16.8/75.3/91.0 μ s. Contract validation passed 453 descriptor and update-packet checks. These are Python contract measurements, not Unreal frame time.

A preliminary link-budget model separates byte accounting from radio-link validation: with no-op packets suppressed, measured SPPA JSON update traffic was 25.8–37.4 kB/s (78.5–94.0 kB/s emitting every packet), between a modeled binary box stream (about 3.27 kB/s) and modeled compressed video (250/625/1250 kB/s at 2/5/10 Mbps).

An adversarial resolver audit over 12 compound or misleading labels forced visual-artifact and equipment-only contexts (toy truck, vehicle shadow, dog statue, person poster, worker helmet) to the unknown fallback, while compound family labels resolved to reviewed archetypes. A complementary offline recipe-review gate accepted dump truck as a reviewed heavy-vehicle mapping, allowed `unlisted obstacle` only as fallback, and rejected `person poster`, `delivery drone`, and `construction crane` for false semantics, unreviewed roles, or runtime-LLM dependence.

Unreal tests exercised the native backend in three steps: an actor-level regression over a real SPPA-DESC-0.2 fixture (12 parts, material and uncertainty tags, pose updates accepted, mismatched or invalid packets rejected), a component replay using the `/api/ui/data` payload shape, and a local-loopback HTTP replay, followed by Editor-Cmd and packaged recorded-payload runs through the same component entry point. This remains an internal short-stream replay, not live telemetry, VR, or operator validation.

Table S10: Selected Unreal evidence that constrains the SPPA claim. Values are preliminary and do not include VR headset, live-network, broad curated-asset validation, or phase-aligned GPU-profiler validation. Auxiliary regression checks are kept in the artifact package, not in this summary table.

Test	Result	Main boundary	Claim impact
Backend switch smoke	Same obstacle input selects assets, actor proxies, or HISM proxies; default remains assets	Editor smoke, not operator workflow	SPPA is optional, not a forced replacement
Editor-Cmd recorded replay	27/27 payload applications accepted across asset, actor-proxy, and HISM backends; SPPA emit-all P95 apply: 1.614/3.805/4.049 ms	component ingestion under Editor-Cmd/Null-RHI	recorded streams reach all backend routes
Packaged recorded replay 20260703T024800Z	27/27 payload applications accepted; frame P95 assets/proxy/HISM: 16.551/6.531/4.809 ms; apply P95: 2.566/5.670/3.246 ms	one short replay, no live network, no phase-aligned GPU profiler, no headset	recorded streams run in packaged executable
Packaged synthetic 100-object replay	actor-proxy pose/shape frame P95: 17.814/19.256 ms	synthetic desktop obstacles; misses 60 FPS P95	desktop feasibility only, not VR readiness
Packaged HISM scaling 20260703T033655Z/033840Z	11 active draw groups; 100-object no-CSV rerun shape frame P95: 37.126 ms; 250/500 shape P95: 94.018/186.113 ms	synthetic desktop obstacles; shape updates miss 30 FPS P95 at 100+ objects	batching helps draw pressure, but dense shape updates remain limiting
Packaged HISM partial updates 20260703T040602Z	10% pose and 10% shape changed-track schedule; 500-object pose/shape frame P95: 16.528/29.787 ms with 11 draw groups	synthetic scheduler rate, not flight-derived; dense all-object updates still fail	partial-update contract supports changed-only scheduling, not dense all-track updates
Packaged cow/tower asset subset	project assets vs. SPPA shape frame P95: 6.789 vs. 22.445 ms	narrow two-label subset	curated assets remain better when available

The packaged evidence consists of synthetic density replay, HISM batching diagnostics, and a short recorded-payload replay. The baseline packaged replay, 20260702T122743Z, used three repetitions, 30 warmup frames, and 120 measured frames per phase. At 100 synthetic obstacles, `semantic_proxy` used 432 components, about 208,864 estimated triangles, and 432 estimated draw calls; pose and shape frame P95 were 17.814 ms and 19.256 ms, passing a 33.3 ms P95 desktop frame budget but missing a 16.7 ms P95 budget; phase-aligned GPU-profiler counters were not captured. The recorded-payload run 20260703T024800Z replayed three externally generated JSONL streams (common asset-compatible, SPPA emit-all, and SPPA changed-only) through the same backend switch; all 27 applications were accepted, with the HISM route at median 6 active draw groups and 4.809 ms frame P95 against 31 components and 6.531 ms for the actor proxy.

Across the HISM sequence, batching reduced the 100-object component and estimated draw-call count from 432 to 11 active draw groups. Dense shape updates are the remaining limit: even after batching, 100-object shape-stream frame P95 stayed above the 30 FPS budget (37.126 ms in the isolated no-CSV rerun) and rose to 94.018/186.113 ms at 250/500 objects. The HISM path was therefore extended with an explicit `partial_obstacle_update` contract—unmentioned entities are retained, mentioned entities replace their old instance keys before new parts are upserted, and the backend verifier checks 6/6 expected live/HISM instances when only one of two entities is updated. Under a changed-track schedule with pose and shape updates each limited to 10% of objects, 500-object pose/shape frame P95 was 16.528/29.787 ms with 11 draw groups and no measured hitches above 33 ms. Stable dense shape-update

performance is not demonstrated even at 100 objects, and the 10% rate is a synthetic scheduler setting rather than a flight-derived one.

A second packaged run adds a narrower real-asset baseline for the two labels with reviewed project static meshes in the repository, `cow` and `tower`. At 50 objects, `project_assets` used 50 static-mesh components and about 167,300 estimated triangles; `semantic_proxy` used 200 components and about 113,200 estimated triangles, with shape-stream P95 6.789 ms versus 22.445 ms for SPPA. Curated assets are preferable when an appropriate asset exists and loads cheaply; SPPA’s intended gap is uncovered or mismatched semantic telemetry, not replacement of available high-quality assets.

S.9 Sealed and Post-Hoc Analysis Tables

Full tables behind the figures of Sections 4.1–4.5 of the main paper. Tables S11 and S12 belong to the confirmatory run; the rest are post-hoc exploratory analyses on the released artifacts and change nothing about H1 or its interval.

Table S11: Method means and paired differences under the prespecified observation conditions, evaluated with the confirmatory run and reported here in full ($n=240$ per condition). The prespecified $+0.030$ margin is exceeded under every condition. Per-condition means of all eight methods are in Table S18.

Condition	SPPA-MVFit	Generic-MVFit	Δ paired	95% CI
Clean	0.557	0.367	0.190	[0.181, 0.199]
Mild morph.	0.512	0.349	0.163	[0.152, 0.173]
Moderate morph.	0.418	0.299	0.118	[0.106, 0.131]
Partial occl.	0.545	0.355	0.189	[0.179, 0.200]
Mask corrupt.	0.555	0.367	0.189	[0.179, 0.197]

Table S12: Paired clean-mask mean IoU differences of SPPA-MVFit minus Generic-MVFit by morphology family and source stratum. All twelve cells are positive; the smallest gain is the compact-vehicle implicit-OOD cell. The equal-stratum aggregate is the primary estimate of Table 1.

Family	CSG-ID Δ	Implicit-OOD Δ	Mean Δ
articulated_vehicle	0.113	0.125	0.119
branching_vertical	0.200	0.210	0.205
compact_vehicle	0.101	0.043	0.072
lattice_tower	0.175	0.163	0.169
quadruped	0.205	0.189	0.197
rider_cycle	0.458	0.299	0.379
Equal-stratum aggregate	0.209	0.172	0.190

Table S13: Graph-by-fitting 2×2 decomposition (clean condition, $n=240$, mean voxel IoU). The Generic no-fit cell was computed after the confirmatory run through the same builder and evaluation, validated bit-exactly against the confirmatory metrics; the remaining cells match confirmatory values. Fitting-effect and per-stratum estimates are in Table S19.

Graph prior	No fit (θ_0)	MVFit	Fitting effect
Generic graph	0.180	0.367	0.187
SPPA family graph	0.427	0.557	0.130
Graph effect (SPPA – Generic)	0.248	0.190	−0.058

Table S14: Drop-one-family re-analysis of the primary paired difference (SPPA-MVFit minus Generic-MVFit, clean condition). Every leave-one-out aggregate stays far above the prespecified $+0.030$ margin.

Excluded family	Δ IoU	95% CI	n
None (all families)	0.190	[0.181, 0.199]	240
compact_vehicle	0.214	[0.203, 0.224]	200
articulated_vehicle	0.204	[0.194, 0.215]	200
quadruped	0.189	[0.178, 0.199]	200
branching_vertical	0.187	[0.178, 0.196]	200
lattice_tower	0.194	[0.184, 0.204]	200
rider_cycle	0.152	[0.143, 0.162]	200

References

- [1] Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. Tripotr: Fast 3d object reconstruction from a single image. *arXiv preprint arXiv:2403.02151*, 2024.
- [2] Zibo Zhao, Zeqiang Lai, Qingxiang Lin, Yunfei Zhao, Haolin Liu, et al. Hunyuan3d 2.0: Scaling diffusion models for high resolution textured 3d assets generation. *arXiv preprint arXiv:2501.12202*, 2025.

Table S15: Wrong-family analysis (clean condition, $n=240$, mean voxel IoU). Rows: true family; columns: family token supplied to the fitter. Bold diagonal: correct token (= confirmatory SPPA-MVFit). Last row: mean over the five wrong tokens per column.

True family ↓ / token →	compact_vehicle	articulated_vehicle	quadruped	branching_vertical	lattice_tower	rider_cycle
compact_vehicle	0.716	0.560	0.462	0.063	0.165	0.213
articulated_vehicle	0.581	0.590	0.386	0.177	0.124	0.167
quadruped	0.349	0.269	0.662	0.281	0.155	0.139
branching_vertical	0.154	0.182	0.299	0.533	0.195	0.134
lattice_tower	0.127	0.084	0.117	0.164	0.313	0.134
rider_cycle	0.116	0.091	0.115	0.060	0.084	0.530
Wrong-token row mean	0.292	0.287	0.239	0.193	0.125	0.093

Table S16: View ablation of SPPA-MVFit (clean condition, $n=240$, correct family token). Single-view arms use an isotropic prior for the unobserved scale, documented before measurement.

Fit variant	Mean IoU	Δ vs dual	CI95 low	CI95 high
Dual-view (sealed SPPA-MVFit)	0.557	—	—	—
Top-only fit	0.458	-0.100	-0.112	-0.088
Side-only fit	0.545	-0.012	-0.021	-0.003
Stratum csg_id: dual 0.551, top-only 0.424, side-only 0.551				
Stratum implicit_ood: dual 0.564, top-only 0.491, side-only 0.540				

Table S17: Generic-graph sensitivity (clean condition, $n=240$, mean voxel IoU). Variants registered in writing before measurement; G1 re-runs the original generic and is validated bit-exactly against the confirmatory metrics. Gap closed is relative to the SPPA=100% endpoint.

Generic graph variant	Mean IoU	Δ SPPA – variant	CI95	Gap closed
G1 sealed generic (control)	0.367	0.190	—	—
G2 box/cylinder chassis	0.339	0.219	[0.210, 0.228]	-15.2%
G3 vertical stack + legs	0.282	0.275	[0.266, 0.285]	-45.0%
G4 slot-wise family mean	0.330	0.228	[0.219, 0.236]	-19.8%
SPPA-MVFit (sealed)	0.557	0.000	—	100.0%

Table S18: All-method means under the prespecified observation conditions (evaluated with the confirmatory run; $n=240$ per condition).

Method	Clean	Mild morph.	Moderate morph.	Partial occl.	Mask corrupt.
SPPA-MVFit	0.557	0.512	0.418	0.545	0.555
Generic-MVFit	0.367	0.349	0.299	0.355	0.367
SPPA text-only	0.427	0.427	0.427	0.427	0.427
Axis-aligned box	0.248	0.264	0.264	0.258	0.248
Ellipsoid	0.342	0.343	0.324	0.353	0.342
Capsule	0.325	0.334	0.316	0.336	0.325
Billboard	0.173	0.153	0.122	0.148	0.170
Visual hull	0.522	0.470	0.356	0.397	0.510

Table S19: Post-hoc exploratory effect estimates for the graph-by-fitting 2×2 decomposition (clean condition; pooled and per stratum).

Effect	Estimate	95% CI	n
Graph effect at no-fit	0.248	[0.234, 0.261]	240
Fitting effect, generic graph	0.187	[0.182, 0.193]	240
Fitting effect, SPPA family graph	0.130	[0.117, 0.143]	240
Total (SPPA-fit – Generic-fit)	0.190	[0.181, 0.199]	240
Interaction (fitting $\times$ graph)	-0.058	—	240
<i>By stratum</i>			
Graph effect at no-fit (CSG-ID)	0.239	[0.220, 0.258]	120
Fitting effect, generic graph (CSG-ID)	0.177	[0.168, 0.186]	120
Fitting effect, SPPA family graph (CSG-ID)	0.147	[0.124, 0.169]	120
Total (SPPA-fit – Generic-fit) (CSG-ID)	0.209	[0.192, 0.225]	120
Graph effect at no-fit (Implicit-OD)	0.256	[0.238, 0.273]	120
Fitting effect, generic graph (Implicit-OD)	0.197	[0.191, 0.204]	120
Fitting effect, SPPA family graph (Implicit-OD)	0.113	[0.098, 0.128]	120
Total (SPPA-fit – Generic-fit) (Implicit-OD)	0.172	[0.162, 0.181]	120

Table S20: Post-hoc exploratory Chamfer distance by observation condition (all methods, $n=240$); paired differences versus SPPA-MVFit with 95% bootstrap intervals.

Method	Clean	Mild morph.	Moderate morph.	Partial occl.	Mask corrupt.	Δ vs SPPA	95% CI
SPPA-MVFit	0.008	0.009	0.012	0.008	0.008	—	—
Generic-MVFit	0.016	0.017	0.021	0.017	0.016	0.008	[0.008, 0.008]
SPPA text-only	0.012	0.012	0.012	0.012	0.012	0.004	[0.004, 0.005]
Axis-aligned box	0.026	0.026	0.120	0.025	0.026	0.018	[0.018, 0.019]
Ellipsoid	0.018	0.018	0.115	0.018	0.018	0.010	[0.010, 0.011]
Capsule	0.020	0.020	0.116	0.019	0.020	0.012	[0.011, 0.012]
Billboard	0.017	0.018	0.114	0.018	0.019	0.009	[0.009, 0.009]
Visual hull	0.009	0.011	0.111	0.011	0.011	0.001	[0.001, 0.001]

Table S21: Post-hoc exploratory worst SPPA-MVFit cases (clean condition, voxel IoU at 64^3). The five sub-threshold cases (IoU < 0.25) are all `lattice_tower` CSG-ID; see the failure analysis in Section 4.2 of the main paper.

Case	Family	Stratum	IoU	Chamfer	Vol. err
test-csg_id-lattice_tower-017	lattice_tower	csg_id	0.147	0.010	1.364
test-csg_id-lattice_tower-007	lattice_tower	csg_id	0.148	0.010	1.400
test-csg_id-lattice_tower-011	lattice_tower	csg_id	0.173	0.009	1.076
test-csg_id-lattice_tower-018	lattice_tower	csg_id	0.182	0.012	1.244
test-csg_id-lattice_tower-008	lattice_tower	csg_id	0.239	0.008	0.228
test-implicit_ood-lattice_tower-015	lattice_tower	implicit_ood	0.251	0.006	0.379
test-csg_id-lattice_tower-000	lattice_tower	csg_id	0.257	0.007	0.449
test-implicit_ood-lattice_tower-013	lattice_tower	implicit_ood	0.257	0.007	0.610
test-csg_id-lattice_tower-005	lattice_tower	csg_id	0.261	0.009	0.005
test-implicit_ood-lattice_tower-009	lattice_tower	implicit_ood	0.265	0.008	0.856

Table S22: Post-hoc exploratory optimizer budget sweep (clean condition, $n=240$, correct family token). The adopted budget is 31 candidates; doubling it gains a non-significant +0.002 IoU for twice the wall time.

Candidate budget	Mean IoU	Mean ms	Median ms	p95 ms
11	0.528	3.574	3.537	4.992
21	0.547	7.590	7.540	9.355
31 (sealed)	0.557	12.566	12.393	14.179
61	0.560	24.034	23.642	27.690

Table S23: Post-hoc exploratory wrong-family paired comparisons (clean condition, $n=240$; stratified bootstrap 95% intervals).

Comparison	Mean diff	CI95 low	CI95 high
Wrong-token mean — correct token	-0.353	-0.362	-0.343
Wrong-token mean — Generic-MVFit	-0.162	-0.167	-0.158
Correct token — Generic-MVFit	0.190	0.181	0.199

Table S24: Post-hoc exploratory oriented bounding-box baseline on the primary benchmark ($n=240$, sealed methods marked). The OBB fitted from the given world-metric box stays within 0.004 IoU of the axis-aligned box, so box orientation does not explain the family-graph effect.

Method	Mean IoU	Δ vs OBB	CI95
Axis-aligned box (sealed)	0.248	-0.004	[-0.005, -0.003]
OBB (this work)	0.252	—	—
Visual hull (sealed)	0.522	0.270	[0.264, 0.275]
SPPA-MVFit (sealed)	0.557	0.306	[0.297, 0.314]

Table S25: Post-hoc exploratory role-aware IoU on the CSG-ID stratum ($n=120$ actors, 920 matched slot-component pairs; slot-to-component mapping frozen in writing before computation). SPPA-MVFit reaches 0.319 role IoU versus 0.017 (cyclic) and 0.053 (random) shuffle controls.

Family	n pairs	Role coverage	Role IoU	Shuffle IoU (cyclic)	Shuffle IoU (random)
compact_vehicle	140	0.724	0.413	0.002	0.062
articulated_vehicle	160	0.614	0.285	0.013	0.046
quadruped	160	0.707	0.423	0.001	0.062
branching_vertical	140	0.304	0.187	0.078	0.066
lattice_tower	160	0.344	0.231	0.004	0.032
rider_cycle	160	0.569	0.372	0.006	0.052
Overall (csg_id)	920	0.545	0.319	0.017	0.053

Table S26: Post-hoc exploratory adversarial-family stratum ($n=120$; twenty actors per family, twelve documented violation types, design frozen before measurement). Clean columns re-run the same actors unviolated. Violating the structural prior cuts the paired SPPA advantage from $+0.209$ to $+0.141$ and produces statistical ties (leaning lattice, cascade crown); SPPA loses at actor level in 8.3% of cases.

Family	n	SPPA clean	SPPA adv	Gen clean	Gen adv	Δ_{adv}	CI95	SPPA < Gen
compact_vehicle	20	0.686	0.481	0.585	0.426	0.055	[0.027, 0.079]	10%
articulated_vehicle	20	0.513	0.465	0.400	0.370	0.095	[0.079, 0.111]	0%
quadruped	20	0.652	0.537	0.447	0.386	0.151	[0.124, 0.178]	0%
branching_vertical	20	0.547	0.343	0.348	0.246	0.098	[0.052, 0.145]	20%
lattice_tower	20	0.316	0.211	0.141	0.088	0.123	[0.067, 0.185]	20%
rider_cycle	20	0.591	0.464	0.134	0.141	0.323	[0.282, 0.362]	0%
Overall	120	0.551	0.417	0.342	0.276	0.141	[0.125, 0.157]	8%

Table S27: Post-hoc exploratory part-query task on the CSG-ID stratum (887 scorable role queries over 120 actors; query set, hull heuristics, and mapping frozen before measurement). F1 and normalized part-centroid error d_c by family. The hull heuristics cover but cannot select; SPPA’s structural count is exact in 70% of actors versus 0% for the hull.

Family	n	SPPA F1	Gen F1	HullA F1	HullB F1	SPPA d_c	Gen d_c	HullA d_c	HullB d_c
compact_vehicle	140	0.555	0.246	0.139	0.123	0.038	0.088	0.272	0.277
articulated_vehicle	160	0.412	0.171	0.140	0.150	0.058	0.134	0.286	0.231
quadruped	143	0.598	0.238	0.123	0.064	0.037	0.082	0.265	0.280
branching_vertical	138	0.253	0.032	0.132	0.044	0.125	0.275	0.151	0.268
lattice_tower	156	0.308	0.037	0.086	0.092	0.047	0.367	0.150	0.231
rider_cycle	150	0.484	0.153	0.049	0.043	0.032	0.106	0.203	0.215
Overall	887	0.434	0.145	0.111	0.087	0.055	0.177	0.221	0.249

Table S28: Real-stream token arm (SPPA-MVFit only; exploratory post-hoc): the 138 GT-matched cases where the real detector emitted a wrong family token (114 cow→tower, 24 biker→tower). Refitting with the correct token collapses the 2D reprojection fit—the correct prior cannot explain the misleading evidence that triggered the wrong label.

Token condition	n	2D reproj. IoU med. [P25, P75]
Real detector token (wrong)	138	0.381 [0.303, 0.410]
Correct-token refit (counterfactual)	138	0.025 [0.023, 0.034]